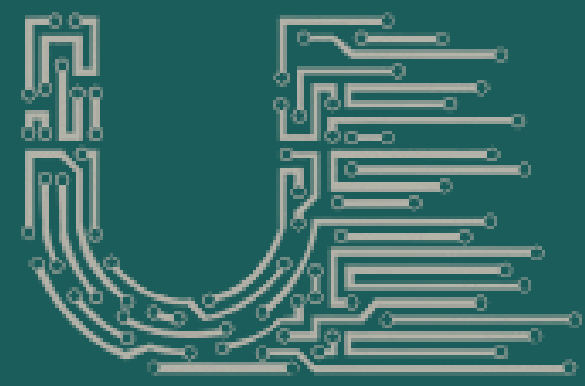




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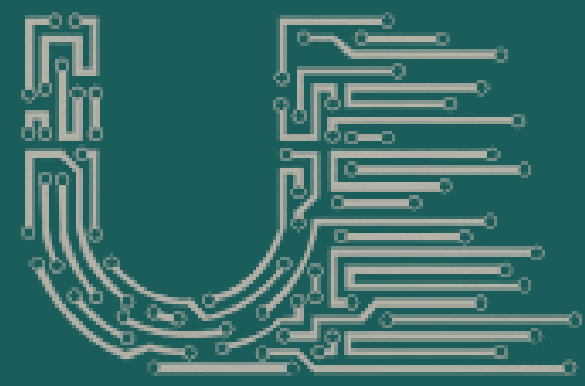
PRESENTS

CONTINUUM PS 2

**Design of Filtering and Shielding
Techniques for CAN Communication
in Electrically Noisy Environments**



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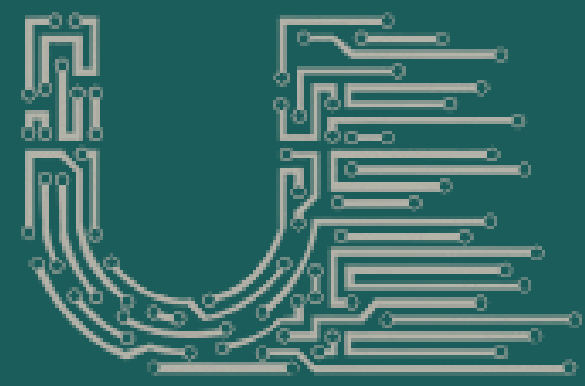
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GENERAL IDEA & PURPOSE

To study, analyze, and design methods for improving the CAN bus communication reliability in environments affected by electromagnetic interference generated by nearby power electronic systems



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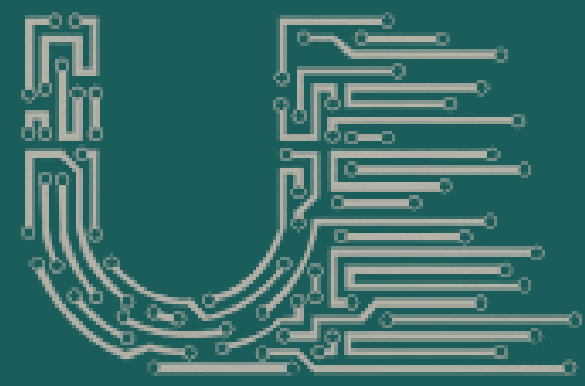
DETAILED PROBLEM STATEMENT

In electric vehicle and power electronic systems, CAN communication lines are often routed in close proximity to high-power switching circuits such as motor controllers, DC-DC converters, and onboard chargers. These circuits generate **high dV/dt and dI/dt** transients, leading to electromagnetic interference through:

- Conducted coupling via shared supply and ground paths
- Capacitive coupling due to high voltage switching nodes
- Inductive coupling due to loop currents and magnetic fields



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DETAILED PROBLEM STATEMENT

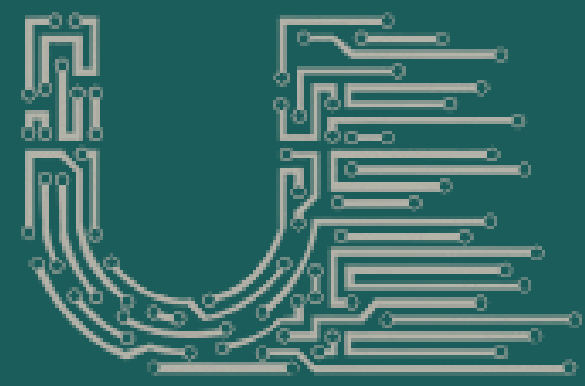
These interference mechanisms introduce:

- Distortion in CAN differential signals
- Increased common-mode noise
- Timing jitter and edge degradation
- Bit errors and communication instability

The problem is to **systematically identify these coupling mechanisms**, quantify their impact on CAN communication, and design **practical filtering and shielding techniques** that improve signal integrity and communication reliability under such conditions.



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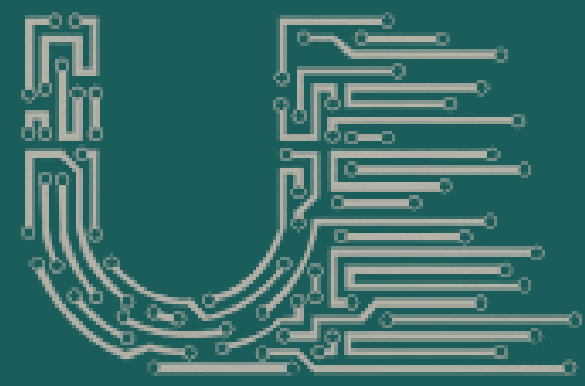
OBJECTIVES

We expect that you systematically:

- Identify and model noise coupling mechanisms affecting CAN signals
- Analyze the interaction between high-power switching circuits and low-voltage communication lines
- Design filtering and shielding techniques for improved signal integrity
- Experimentally or analytically validate improvements in communication reliability*
- Establish quantitative performance metrics for CAN communication in noisy environments



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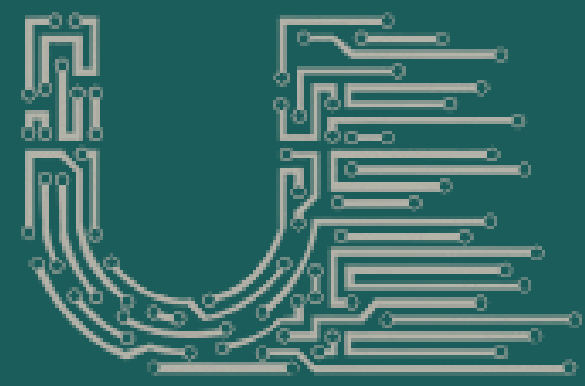
ASSUMPTIONS

To arrive at a solution, it is assumed that:

- CAN protocol parameters follow standard specifications (bit rate, termination, signaling levels, etc.)
- Cable length and termination are within recommended limits
- Noise sources are primarily from nearby power electronic switching circuits
- Environmental conditions such as temperature and humidity are controlled
- Measurement tools (oscilloscope, CAN analyzer, etc.) are available or can be simulated*



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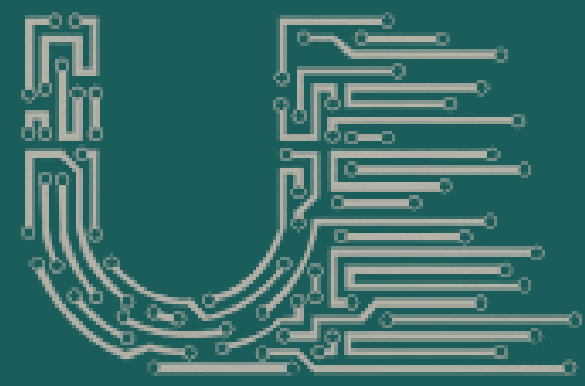
SCOPE OF WORK

The participants' solutions shall contain:

- Identification and modeling of noise sources and coupling paths
- Baseline evaluation of CAN signal integrity under noisy conditions
- Design and implementation of filtering techniques (passive components, impedance control)
- Design and evaluation of shielding and grounding strategies
- Comparative analysis between baseline and improved designs



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TARGET PERFORMANCE PARAMETERS

Communication Reliability

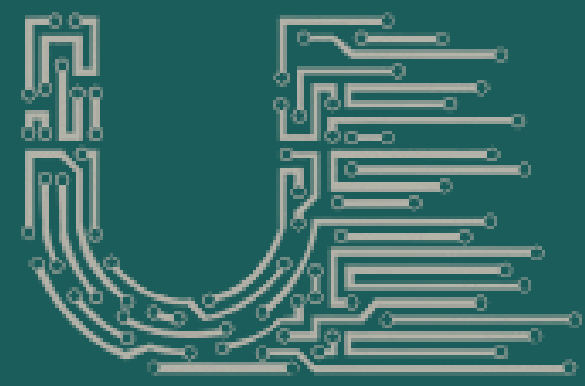
- Bit error rate (BER): $\leq 10^{-9}$
- No frame loss during continuous operation under switching conditions

Signal Integrity

- Differential impedance: $120\Omega \pm 10\%$ (measured or controlled through design)
- Differential signal amplitude maintained within CAN standard limits
- Signal rise/fall time degradation: $\leq 20\%$ **compared to ideal** waveform



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TARGET PERFORMANCE PARAMETERS

Noise Metrics

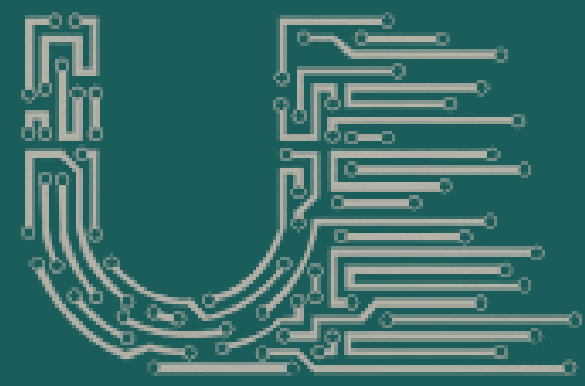
- Common-mode noise on CAN lines: **≤ 100 mV (peak)**
- Conducted noise coupling from power circuits: **≤ 50 mV (peak)**
- Reduction in noise after mitigation techniques: **$\geq 50\%$ compared to baseline**

Robustness

- Stable communication during:
 - Motor switching events
 - DC-DC converter full-load operation
- No communication reset or bus-off condition during testing



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EXPECTED DELIVERABLES

1. System Definition and Baseline Setup:

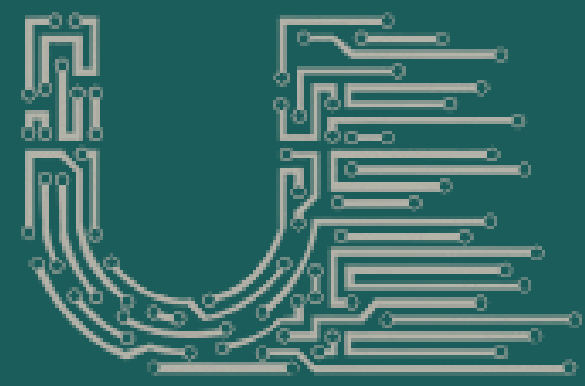
- Description of system under study (block diagram)
- Identification of noise sources and coupling paths
- Baseline measurement or simulation of CAN signals without mitigation

2. Problem Quantification:

- Waveform capture or simulation showing:
 - Signal distortion
 - Noise injection
 - Error occurrence (if any)



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EXPECTED DELIVERABLES

2. Problem Quantification:

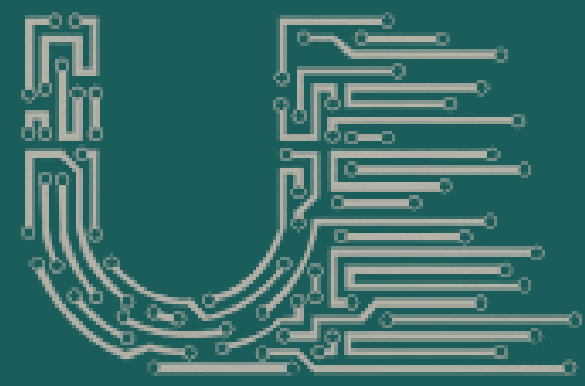
- Quantitative baseline values
 - Noise levels
 - Signal amplitude
 - Communication errors

3. Design of Mitigation Techniques

- At least two different approaches, such as:
 - Passive filtering (RC / LC / common-mode choke)
 - Shielding (cable shielding, enclosure strategy)
 - Grounding improvements
 - Routing and layout changes



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EXPECTED DELIVERABLES

3. Design of Mitigation Techniques:

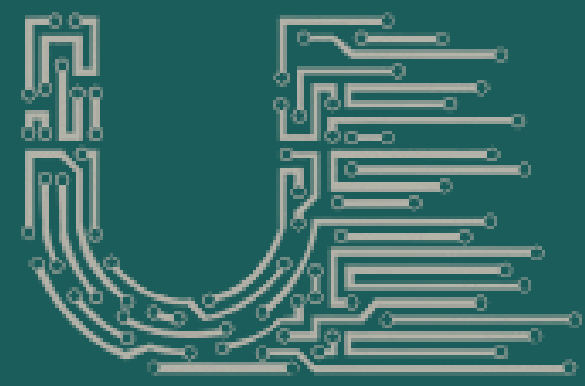
- Design justification
 - Why this method
 - Expected impact
 - Design trade-offs

4. Comparative Analysis:

- Performance comparison between
 - Baseline (no mitigation)
 - Technique 1
 - Technique 2
- Parameters to compare
 - Noise levels
 - Signal waveform quality
 - Communication reliability



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EXPECTED DELIVERABLES

5. Validation Results:

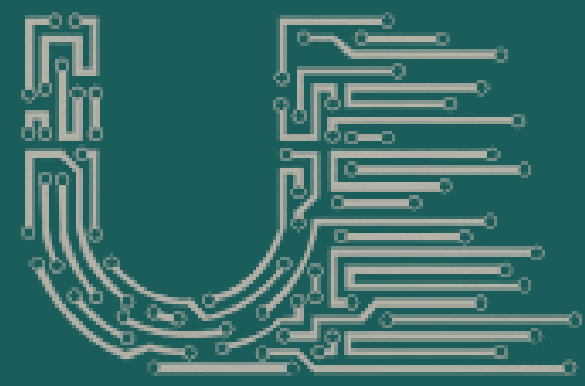
- Measured or simulated results demonstrating
 - Noise reduction
 - Improved signal integrity
 - Improved communication stability

6. Critical Discussion:

- Limitations of implemented techniques
- Practical challenges in real systems
- Scalability and applicability to larger EV systems



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PRIZE POOL

1st Place → ₹ 10,000

2nd Place → ₹ 7,500

3rd Place → ₹ 5,000

Separate goodies for freshers!

DEADLINE: 12 APR 2026

ALL THE BEST!



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